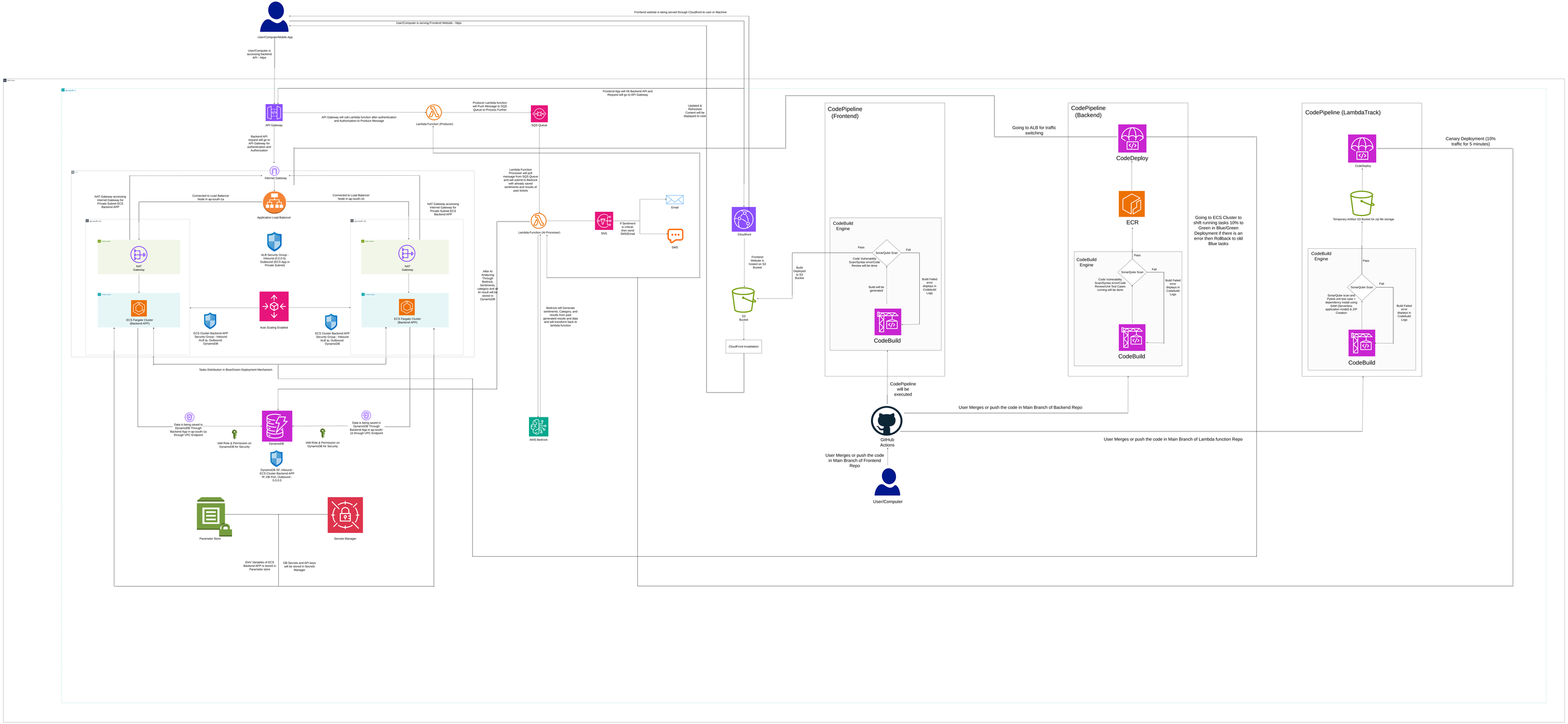

User/Computer/Mobile App
Frontend website is being served through Cloudfront to user or Machine
User/Computer is serving Frontend Website - https
User/Computer is accessing backend API - https
AWS Cloud
ap-south-1
Frontend App will Hit Backend API and Request will go to API Gateway
Updated & Refreshed Content will be displayed to User
API Gateway
API Gateway will call Lambda function after authentication and Authorisation to Produce Message
Lambda Function (Producer)
Producer Lambda function will Push Message to SQS Queue to Process Further
SQS Queue
Backend API request will go to API Gateway for authentication and Authorization
VPC
Internet Gateway
NAT Gateway accessing Internet Gateway for Private Subnet ECS Backend APP
Connected to Load Balancer Node in ap-south-1a
Application Load Balancer
Connected to Load Balancer Node in ap-south-1b
NAT Gateway accessing Internet Gateway for Private Subnet ECS Backend APP
ap-south-1a
Public subnet
NAT Gateway
Private subnet
ECS Fargate Cluster (Backend APP)
ALB Security Group - Inbound (0.0.0.0), Outbound (ECS App in Private Subnet)
Auto Scaling Enabled
ECS Cluster Backend APP Security Group - Inbound ALB Ip, Outbound DynamoDB
ap-south-1b
Public subnet
NAT Gateway
Private subnet
ECS Fargate Cluster (Backend APP)
ECS Cluster Backend APP Security Group - Inbound ALB Ip, Outbound DynamoDB
Tasks Distribution in Blue/Green Deployment Mechanism
Lambda Function Processer will poll message from SQS Queue and will submit to Bedrock with already saved sentiments and results of past tickets
Lambda Function (AI Processer)
SNS
If Sentiment is critical, then send SMS/Email
Email
SMS
After AI Analyzing Through Bedrock, Sentiments, category and all AI result will be stored in DynamoDB
Bedrock will Generate sentiments, Category, and results from past generated results and data and will transform back to lambda function
AWS Bedrock
Data is being saved in DynamoDB Through Backend App in ap-south-1a through VPC Endpoint
IAM Role & Permission on DynamoDB for Security
DynamoDB
IAM Role & Permission on DynamoDB for Security
Data is being saved in DynamoDB Through Backend App in ap-south-1b through VPC Endpoint
DynamoDB SF, Inbound - ECS Cluster Backend APP IP, DB Port, Outbound - 0.0.0.0
Parameter Store
Secrets Manager
ENV Variables of ECS Backend APP is stored in Parameter store
DB Secrets and API keys will be stored in Secrets Manager
Cloudfront
Frontend Website is hosted on S3 Bucket
S3 Bucket
CloudFront Invalidation
Build Deployed to S3 Bucket
CodePipeline (Frontend)
CodeBuild Engine
Pass
SonarQube Scan
Fail
Code Vulnerability Scan/Syntax error/Code Review will be done
Build will be generated
Build Failed error displays in Codebuild Logs
CodeBuild
CodePipeline will be executed
GitHub Actions
User Merges or push the code in Main Branch of Frontend Repo
User/Computer
User Merges or push the code in Main Branch of Backend Repo
User Merges or push the code in Main Branch of Lambda function Repo
Going to ALB for traffic switching
CodePipeline (Backend)
CodeDeploy
ECR
CodeBuild Engine
Pass
SonarQube Scan
Fail
Code Vulnerability Scan/Syntax error/Code Review/Unit Test Cases running will be done
Build Failed error displays in Codebuild Logs
CodeBuild
Going to ECS Cluster to shift running tasks 10% to Green in Blue/Green Deployment if there is an error then Rollback to old Blue tasks
CodePipeline (LambdaTrack)
CodeDeploy
Temporary Artifact S3 Bucket for zip file storage
CodeBuild Engine
Pass
SonarQube Scan
Fail
SonarQube scan and Pytest unit test case + dependency install using SAM (Serverless application model) & ZIP Creation
Build Failed error displays in Codebuild Logs
CodeBuild
Canary Deployment (10% traffic for 5 minutes)